Sarang Manoj Pekhale

PhD Research Candidate, CSCCM Lab, Department of Applied Mechanics, Indian Institute of Technology, Delhi

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My research aims to develop **Agentic AI Systems**, bringing multimodal reasoning into **Computational Science** workflows, drawing inspiration from how human scientist observes, hypothesize, and validate. This will hopefully extend the human scientific capabilities to explore wide range of new, valid, and feasible hypotheses. Fundamentally, I will be focusing on three coupled domains, **Discovery**, answering *"what equations governs this physical system ?"*, **Analysis**, finding *"what does the mathematical model predicts ?"*, and **Design**, building *"what configurations achieves the goal ?"*.

Publications

[1] **Sarang Manoj Pekhale**, Amartya Roy, Rajat Sarkar, and Souvik Chakraborty. "See, Hypothesize, Validate: Multimodal Agentic Framework for Discovering Governing PDEs". *Submitted to AAAI Conference 2027 (In Review), arXiv preprint*.

[2] **Sarang Manoj Pekhale**, Tushar, Heinrich Stueer, Stefan Schmitt, Rishabh Shrivastava, Amit Yadav, Souvik Chakraborty. "BRIDGING NUMERICAL SIMULATION AND AI: A HYBRID FRAMEWORK FOR FLUTTER STABILITY ANALYSIS". *Submitted for internal review : Siemens Energy*.

Education

Jan 2025 – present
PhD, Agentic Artificial Intelligence for Computational Science, *CSCCM Lab, Applied Mechanics Department, Indian Institute of Technology*, Delhi, India.
Advisor: Prof. Dr. Souvik Chakraborty.
Semester I: Deep Learning, Optimization; CGPA: 8.571.
Semester II: Reliability-based Design Optimization, Probability and Statistics; CGPA: 8.769.
Cleared comprehensive examination in Aug 2026.

July 2022 – June 2024
M.Tech, Modeling and Simulation, *Department of Scientific Computing, Modeling and Simulation, Savitribai Phule Pune University*, Pune, India.
Advisor: Prof. Dr. Bhalchandra Pujari.
Thesis: Developing a ML approach to accurately identify the size of Impactors/Projectiles responsible for forming Lunar Surface Craters.
Coursework: Probability Theory, Linear Algebra, Real Analysis and Calculus, Vector Calculus, Statistical Inference, Optimization, Ordinary Differential Equations, Modeling and Simulation, Fundamentals of Computing, Numerical Computing.
Electives: Machine Learning, Computational Fluid Dynamics.
Grade: 9.21 CGPA.

July 2018 – June 2022
B.E., Mechanical Engineering, *Progressive Education Society's Modern College of Engineering, Savitribai Phule Pune University*, Pune, India.
Advisor: Prof. Shyam Ramnani.
Thesis: Study of Heat Transfer Enhancement in Radiator using Nanofluid.
Grade: 9.47 CGPA.

Research Experience

Jan 2025 – Present
PhD Researcher, *Siemens Energy*, New Delhi, India.
- **Physics Informed Machine Learning Model for Flutter Predictions.**

July 2024 – Dec 2024
AI Research Intern - Engineering Applications, *Thermax Limited*, Pune, India.
- **Predictive Modeling of Porosity Distribution in Electrostatic Precipitator Screens Using Surrogate-Driven Artificial Neural Networks.** Developed an Artificial Neural Networks model to predict gas distribution screens in electrostatic precipitators (ESP) using CAD-extracted dimensions as features which were used in CFD simulations and scaled target variable i.e. gas distribution screen images to 64x64 pixels, preserving spatial porosity patterns.

Jan 2024 – June 2024
Deep Learning Intern, *Department of Scientific Computing, Modeling and Simulation, Savitribai Phule Pune University*, Pune, India.
Advisor: Prof. Dr. Bhalchandra Pujari.
- **Developing a ML approach to accurately identify the size of Impactors/Projectiles responsible for forming Lunar Surface Craters.** Developed a transfer learning ML framework using ANN to predict the size of the impactors from the lunar surface craters as input data, to identify new outer space elements and to consider prior precautions for events caused by the impactors.

Jan 2022 – June 2022 **Study of Heat Transfer Enhancement in Radiator using Nanofluid**, *Mechanical Engineering Department, Progressive Education Society's Modern College of Engineering, Savitribai Phule Pune University*, Pune, India.

Advisor: Prof. Shyam Ramnani.

– Build a wet lab experimental setup to study the heat transfer enhancement in radiator using metallic nanofluids.

Skills

Programming Autogen, Python, Numpy, Scipy, Pytorch, TensorFlow, Keras, Matplotlib, Seaborn, Pandas, Plotly, R.

Software ANSYS Workbench, ANSYS BladeGen, ANSYS Fluent, ANSYS CFX, ANSYS Mechanical APDL, ANSYS TurboGrid, Design Modeller, Spaceclaim, BladeGen, SolidWorks.

Languages English, Professional; Hindi, Professional; Marathi, Native.

References

[Referee 1] **Prof. Dr. Souvik Chakraborty**
Associate Professor, Applied Mechanics Department, Yardi School of Artificial Intelligence, Indian Institute of Technology, Delhi, India
Advisor
E-mail: souvik@am.iitd.ac.in
CSCCM | Google Scholar | LinkedIn

[Referee 2] **Prof. Dr. Somdatta Goswami**
Assistant Professor, Civil and Systems Engineering, Applied Mathematics and Statistics and Data Science and AI Institute, Johns Hopkins University, Baltimore, MD, USA
Co-Advisor
E-mail: somdatta@jhu.edu
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